# Magnetic properties of Precambrian pillow lavas of the Mona Complex and a related dyke swarm, Anglesey, Wales

*J. D. A. Piper*

Four palaeomagnetic sites in Monian pillow lavas of the Gwna Group from Llanddwyn Island, Anglesey, yield statistically-significant groups of magnetic directions after demagnetising in low fields. A palaeomagnetic pole of 57°E, 16½°S is derived after dip correction assuming no plunge components, and is believed to apply to the time of formation of the pillow lavas (*ca.* 600 m.y.). By contrast to young pillow lavas these rocks have poor magnetic stability and low Köenigsberger ratios although petrologic features are comparable; this is believed to be due to replacement of the finer grain size fractions of the titanomagnetite. Five of the seven dolerite dykes cutting the Monian rocks here yield pole positions close to those of Upper Silurian to Lower Devonian rocks from elsewhere in Britain confirming that they are Palaeozoic in age.

## 1. Introduction

This paper reports a palaeomagnetic investigation of Precambrian pillow lavas of the Mona Complex in Anglesey which are exposed to advantage at Llanddwyn Island on Newborough Warren; the palaeomagnetism of a swarm of dolerite dykes cutting the Monian rocks at this locality is also described. It is normal procedure in palaeomagnetic studies to undertake field tests such as fold or conglomerate tests (Graham 1949; Irving 1964) to evaluate the time at which the magnetisation was acquired, and such tests are of critical importance where there is a possibility that remagnetisation (Creer 1967; Storetvedt 1970) may have taken place. Unfortunately, since diminishing surface exposure generally accompanies increasing geologic age, such tests become difficult to apply to older rocks. On Llanddwyn Island such tests cannot effectively be applied and it is then logical to look for properties of the magnetisation which may relate to mechanisms of formation. Examples of such properties are those related to quenching (Watkins *et al.* 1970; Watkins and Paster 1971) and produced when magma is chilled on eruption into water; the pillow lavas were accordingly examined in this context.

## 2. Geological outline

The Monian rocks of Anglesey comprise a complex supracrustal sequence which includes greywackes, phyllites, quartz-rich sandstones and spilitic pillow lavas in

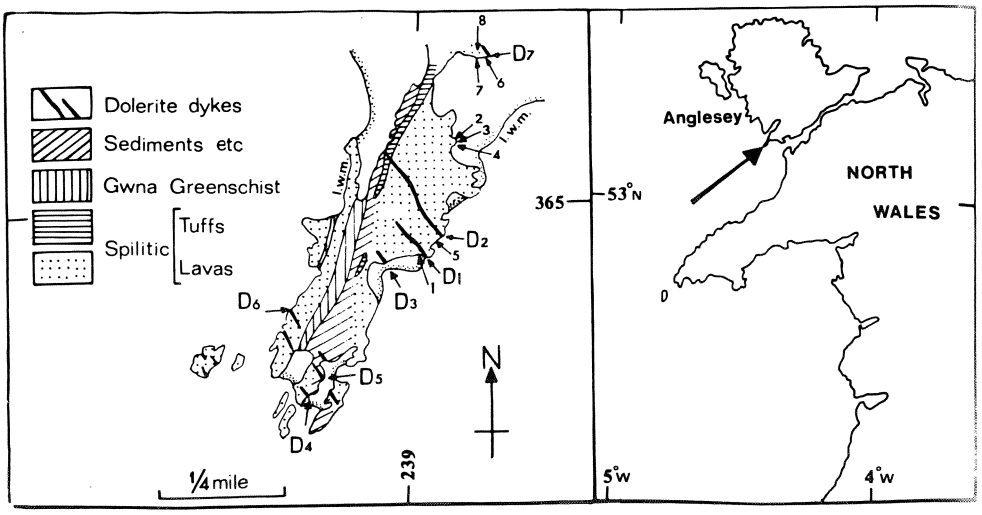
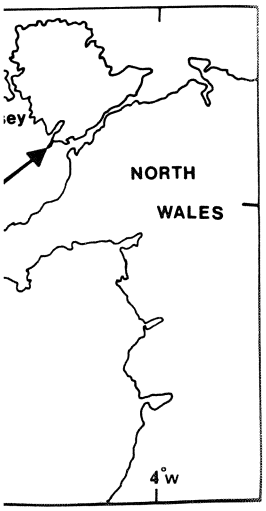



Fig. 1. Simplified geological map of Llanddwyn Island, southwest Anglesey (355·5°E, 53·1°N) showing the locations of palaeomagnetic sampling sites; reference lines are National Grid lines and the coast is defined by low tide level. The small scale map gives the regional location. Lithological divisions are simplified after Greenly (1919).

addition to gneisses whose status is problematical. They have been described in great detail by Greenly (1919, 1923) who recognised six major lithological divisions, and Shackleton (1953, 1954). The latter author gives a structural interpretation which places the pillow lavas studied here (Gwna Group) near the top of the succession as the second youngest of the six divisions recognised by Greenly. Folding and metamorphism took place during a regional isotopic event dated 580–600 m.y. by Moorbath and Shackleton (1966). A K–Ar (muscovite) age of 611 m.y. has also been given for a schist from the Gwna Group (Fitch *et al.* 1969). These radiometric results date a metamorphic episode believed by Shackleton (1954) to be a regional event; limited exposures of glaucophane schist occur within the Monian outcrop but the pillow lavas at Newborough Warren are practically unmetamorphosed (Moorbath and Shackleton 1966). On Anglesey, the Monian rocks are unconformably overlain by the volcanic rocks of the Arvonian Group which, in North Wales, pass conformably into Lower Cambrian sediments.

At Llanddwyn Island a suite of NW–SE trending dolerite dykes was intruded into the Monian rocks subsequent to their folding. These are parallel to dykes of established Tertiary age elsewhere in Anglesey (Greenly 1919; Dagley 1969; Evans *et al.* 1973) but at Newborough Warren they were regarded as Palaeozoic by Greenly (1919) although there is no definitive evidence for this.

Palaeomagnetic samples were drilled from pillow lavas at eight localities (Fig. 1) and were taken at measured distances across individual pillows. Each of the seven dolerite dykes cutting the Monian rocks was also drilled at one locality. Up to eight samples oriented by sunsights were taken from each locality, and were cut into one-inch long cylinders for palaeomagnetic measurement.




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## 3. Palaeomagnetic results

All samples were measured with a parastatic magnetometer and progressively demagnetised by the alternating field (a.f.) method in steps of 50 or 100 Oersteds up to peak fields between 500 and 1400 Oersteds. Directional behaviour and demagnetisation characterististics of typical samples of the pillow lavas are shown in Figure 2. All samples with the exception of those from site 5 show only poor stability to a.f. treatment. Directions of magnetisation settle out to no single stable end point, and erratic demagnetisation curves suggest that magnetic components are both introduced and subtracted by the demagnetisation procedure. Comparable behaviour is exhibited by Precambrian Birrimian greenstones of West Africa (Piper and Lomax 1973) where inconclusive tests suggest that the magnetisation is held predominantly by multidomain carriers; it is consistent with low Köenigsberger ratios (Qo, the ratio of remanent to induced magnetisation) which for this collection lie in the range 0·01–0·10.

In spite of this inherent instability most samples retain easterly declinations and shallow inclinations after dip correction. Prior to this dip correction (85° in a direction 147°E) these directions are quite different from the present Earth's magnetic field direction at the sampling locality and are at least not of recent acquisition. Samples from sites 3, 6 and 7 give statistically-significant groupings (Irving 1964, table 4.3) after demagnetising in low fields up to a few hundred Oersteds (Table I). Samples of site 5 contrast to the remainder of the collection by showing stable directional behaviour and systematic fall in magnetic moments with progressive a.f. treatment (Fig. 2). They reach stable end points after demagnetising in low fields with about 10% of initial magnetic moments remaining and a mean direction of D=298°, I=−27° which is approximately antiparallel to the mean directions of sites 3, 6 and 7. This result may imply a reversal of magnetisation within the time of eruption of the Gwana Group pillow lavas, although as discussed in the next section, the magnetic remanence carriers of site 5 are quite different from the remaining sites and the precise significance of this apparent reversal of magnetisation is uncertain.

Taking the antiparallel direction of magnetisation of site 5 and combining with sites 3, 6 and 7 yields a mean direction of D=126·5°,I=+9·1° giving a palaeomagnetic pole at 57·1°E, 16·2°S (Fisher's (1953) precision parameter k=12). The palaeomagnetic directions are corrected for the tectonic tilt of the pillow lavas and it is not known whether plunge components should be taken into consideration here; if they are present they would alter the quoted palaeomagnetic pole position.

## 4. Opaque petrology and thermomagnetic characteristics

All samples examined petrologically (by Reichert Zeto-pan Pol microscope under oil immersion to ×1900 magnification) except those from site 5 show abundant sulphides with grain sizes from about 5$\mu$ down to limits of resolution. Sulphides are typically more abundant than titanomagnetite which occurs as grains, generally of skeletal form, from a few microns in size down to limits of resolution. No deuteric oxidation was found in these sites, although grains are in almost all cases partially replaced at the margins by non-reflecting aggregates. The opaque mineralogy of site 5 is quite different: no sulphides are present and titanomagnetite grains are up to an order larger in size. All grains show some evidence of deuteric oxidation with subdivision by ilmenite lamellae equivalent to oxidation classes 2–3 of Wilson and Watkins (1967); marginal alteration of the grains is still evident, abundant red

Table I. Palaeomagnetic results from pillow lavas of the Gwna Group, Llanddwyn Island, Anglesey (355·5°E, 53·1°N) after a.f. cleaning and correction for tectonic tilt.

| *Site No.* | *Cleaning Field (Oersteds)* | *N* | *D* | *I* | *R* | *k* | $\alpha_{95}$ | *Pole Position* °*E* | °*N* |
|---|---|---|---|---|---|---|---|---|---|
| 3 | 150 | 5 | 122 | −19 | 4·12 | 5 | 40 | 65 | −27 |
| 5 | 100 | 5 | 298 | −27 | 4·90 | 40 | 12 | 234 | 5 |
| 6 | 200 | 6 | 135 | 0 | 4·73 | 4 | 39 | 47 | −25 |
| 7 | 200 | 3 | 129 | 28 | 2·95 | 41 | 19 | 45 | −9 |
| Mean direction and pole position for 4 sites taking the reversed direction of site 5: | | | 126·5 | 9·1 | 3·74 | 12 | 16·2 | 57·1 | −16·5 |

NOTE: N = number of samples; R = length of resultant of N unit vectors; D = declination and I = inclination of remanent magnetisation; k = (N−1)/(N−R) = estimate of Fisher's (1953) k parameter; $\alpha_{95}$ = semi angle of cone of 95% confidence for mean direction.

internal reflections suggesting that these include iron hydroxides, possibly goethite.

Certain characteristics emerge from opaque petrological studies of Tertiary to Recent pillow lavas (Watkins *et al.* 1970; Brooke *et al.* 1970; Lawley and Ade-Hall 1971). In quenched material there is a ubiquitous presence of non-deuterically oxidised titanomagnetite, often skeletal in form, and associated with abundant sulphides. These characteristics are typical of pillow lavas of this study except those of site 5. Absence of sulphides, larger grain size and limited deuteric alteration at this latter site indicate slower cooling under conditions of higher oxygen fugacity (Ade-Hall *et al.* 1968; Lawley and Ade-Hall 1971) during which any sulphur present was lost by oxidation to a gaseous phase; these properties seem to be exceptional for pillow lava material.

At least four thermomagnetic determinations were made from samples of each site and while samples from the same locality generally show similar characteristics, there is considerable variation between sites. Values of saturation magnetisation ($J_S$) were low for all samples except those from site 5 and precise Curie points could not be obtained from all samples. Typical $J_S$-temperature curves are shown in Figure 3.

Fig. 2. Demagnetisation curves and directional behaviour of some palaeomagnetic samples from Gwna Group pillow lavas. The abscissa of the demagnetisation curves are graduated in intervals of 100 oersteds peak demagnetising field and M/Mo is the fraction of the initial magnetic moment remaining after cleaning. Projections are stereographic and directions of magnetisation are dip corrected; open symbols are upper hemisphere and closed symbols are lower hemisphere plots.

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| $\alpha_{95}$ | Pole Position °E | °N |
|---|---|---|
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| 12 | 234 | 5 |
| 39 | 47 | −25 |
| 19 | 45 | −9 |
| 16·2 | 57·1 | −16·5 |

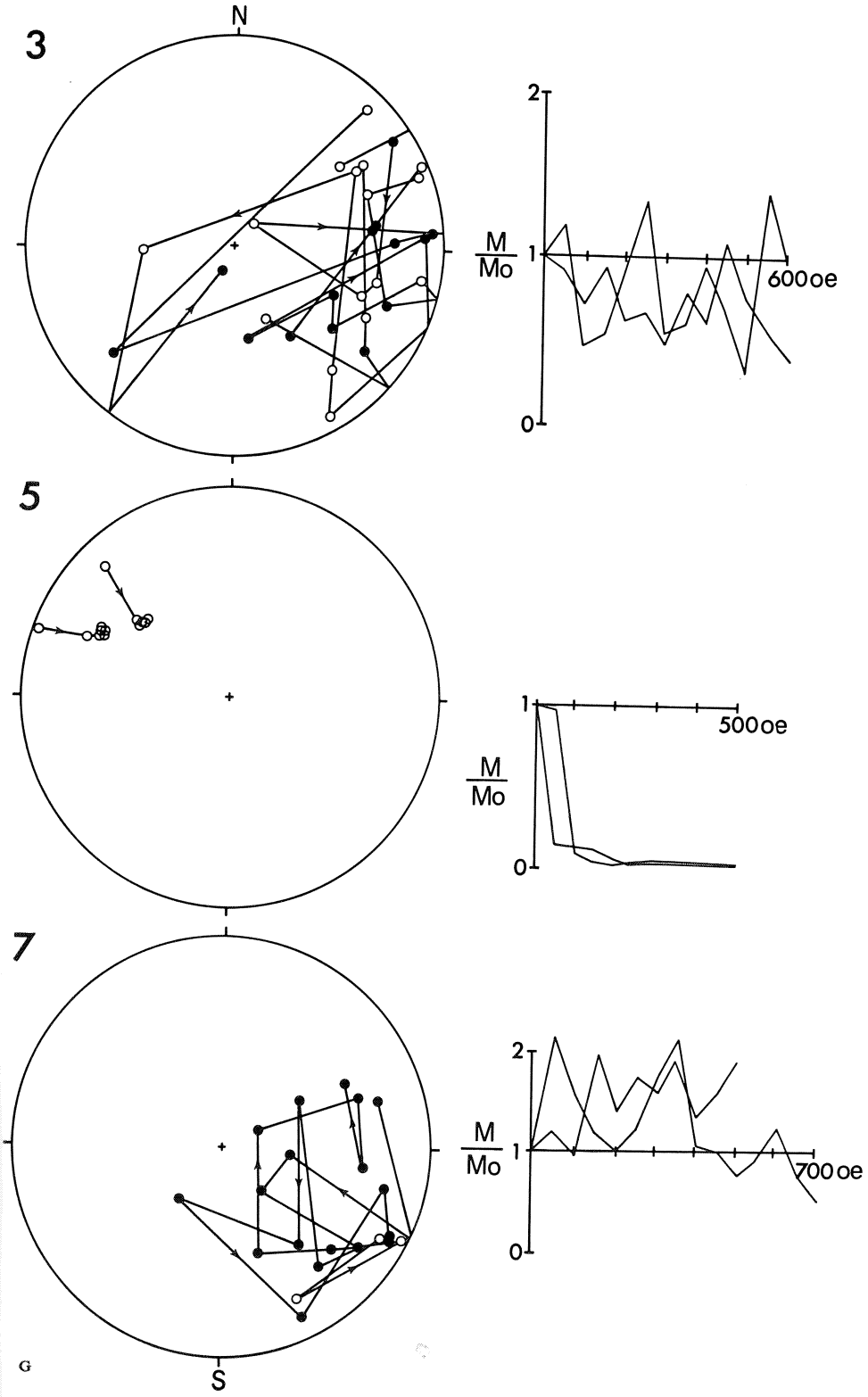
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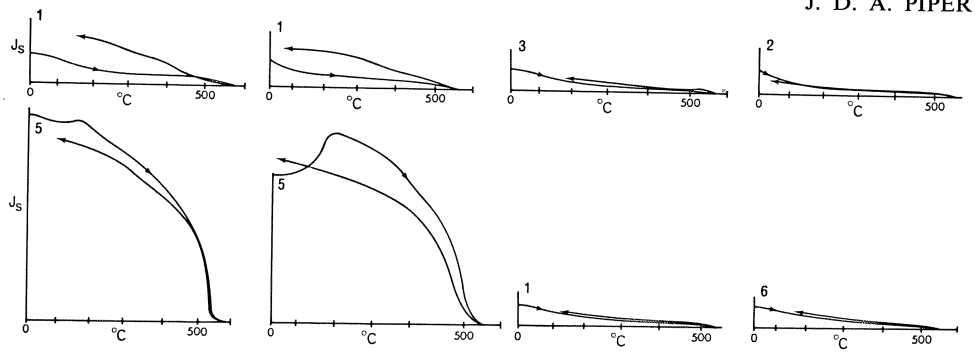
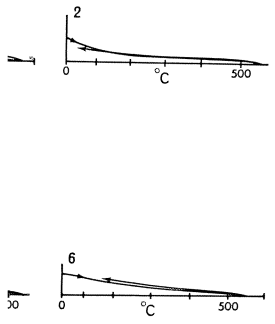



Fig. 3. Typical thermomagnetic curves from palaeomagnetic samples of Gwna Group pillow lavas showing saturation magnetisation (Js) plotted as a function of temperature; the numbers refer to palaeomagnetic sites.

Most curves are irreversible with small to moderate increases in $J_S$ on cooling. Upshifts of the $J_S$-temperature curves (well shown by samples from site 1) might result from low temperature oxidation, formation of ilmenite lamellae or internal ordering (Larsen *et al.* 1968). The cases investigated here almost certainly result from continuous oxidation (unmixing of titanomagnetite and maghemite as reported by Ozima and Ozima (1971)) since there is evidence for a low Curie point on most curves which does not appear on the cooling curve. Presence of maghemite resulting from low temperature oxidation of titanomagnetite is another characteristic of young pillow lavas (Ozima and Larson 1970), although small grain size has precluded positive indentification by optical means in this study.

Samples from site 5 have saturation magnetisations considerably greater than other sites and give single Curie points between 525 and 575°C. Thermomagnetic curves are characteristic of a single magnetic phase and compatible with deuterically-oxidised titanomagnetite observed in polished sections. There is slight decrease in $J_S$ on cooling indicating partial formation of a mineral with lower saturation magnetisation, most probably oxidation to hematite. An interesting feature of site 5 is the continuous increase in $J_S$ to temperatures of about 190°C. This feature is not reproduced by reheating and is sometimes present in thermomagnetic curves of other workers (e.g. Watkins and Cambray 1970) but its explanation is not clear. It must result from transformation of a low temperature phase into a magnetic component which the final Curie point indicates is magnetite; the observation that iron hydroxides are present in samples of site 5 suggest that continuous conversion to magnetite on heating could produce this increase in $J_S$.

Figure 4 shows the spatial distribution of magnetic properties across individual pillows; each traverse terminates close to the centre of the pillow concerned. Although Tertiary and younger pillow lavas show several significant variations in magnetic properties from quenched margins to crystalline interiors (e.g. Watkins *et al.* 1970), there are no comparable systematic changes in properties across these Precambrian pillows. These variations (notably in Köenigsberger ration Qo, and magnetic susceptibility $\chi$) are probably not found for two reasons: firstly, glassy margins of the pillows will devitrify with time to produce high susceptibility and low remanent intensity mineral assemblages. Secondly, since there is normally some variation in the spectra of coercivities across the pillows, possible spontaneous decay of primary moments will be accompanied by acquisition of secondary viscous components of magnetisa-




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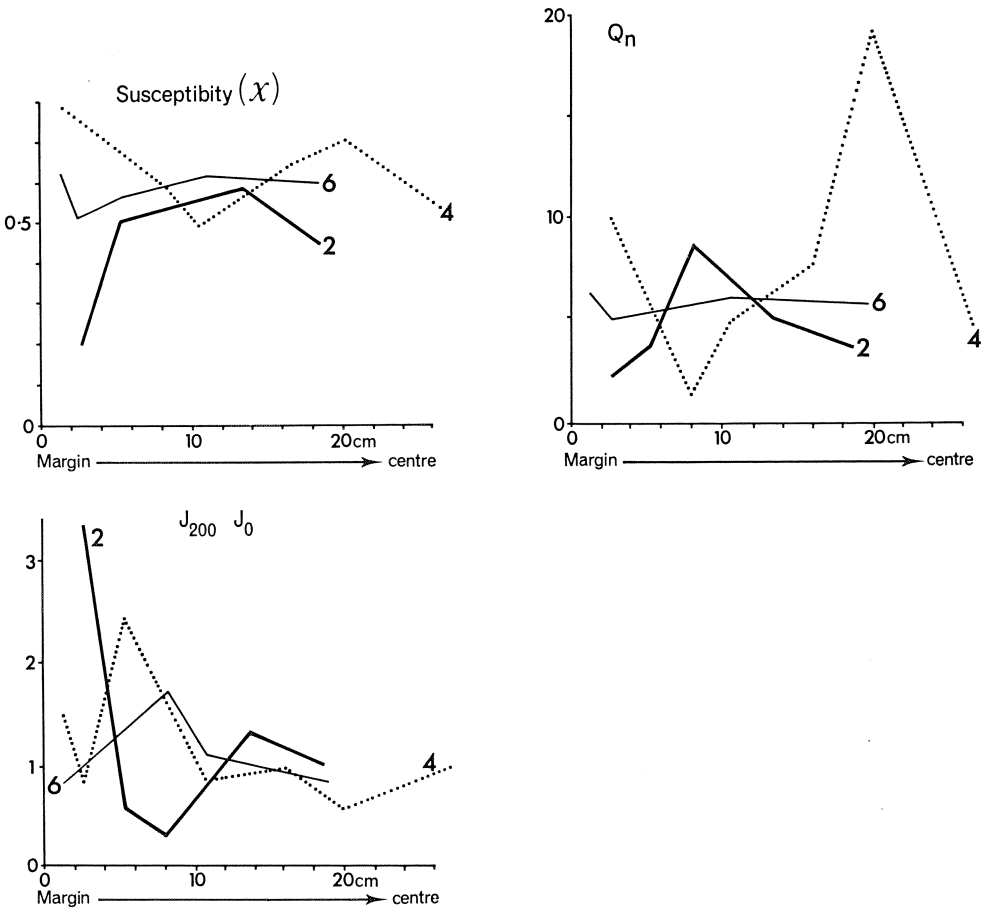



Fig. 4. Variation in some magnetic properties with distance across individual pillows. The magnetic properties plotted here are Köenigsberger ratio (Qo), magnetic susceptibility ($\varkappa$), and $J_{200}/J_0$ (the ratio of intensity of magnetisation after demagnetising in 200 oersteds to initial intensity) as a measure of coercivity.

tion to change the initial moments by varying amounts. Thin section examination of site 6 confirms that about 80% of the marginal sample is composed of low birefringent products resulting from devitrification, and other samples inside of this chilled carapace show little systematic variation in grain size. At this site pyroxene is variably altered to secondary amphibole and feldspars are only slightly sericitised.

Thus, although Monian pillow lavas still possess a number of magnetic and petrologic features reminiscent of young pillows, there are several notable differences. Firstly, absence of the high $Q_n$ values characteristic of young quenched material (Ade-Hall 1964) could result from any or all of three causes: a magnetisation acquired in a weak ambient field, spontaneous decay of magnetisation with time, or acquisition of a low temperature chemical remanent magnetisation (CRM) by oxidation to titanomaghemite (a CRM due to oxidation to a member of the hematite-ilmenite series is not suggested by magnetic or petrologic properties of any samples of this collection). The latter of these three processes appears to be an important one in the present deep ocean floor environment (Ozima, 1971).

Secondly, as appears to be typical of Palaeozoic and older pillow lava material, magnetic moments of the Monian pillows are only poorly-stable in contrast to the high stabilities exhibited by most young pillow material. Magnetic stability in igneous rocks is generally attributed to single or pseudo-single domains produced either by rapid initial cooling or subsequent subdivision of titanomagnetite grains by deuteric oxidation (e.g. Larson *et al.* 1968). Since pillow lavas are generally unoxidised, the magnetic stability of young pillows must be attributable to the former cause. Although optically visible grains ($>0{\cdot}5\mu$) have sizes much larger than single domains ($<0{\cdot}03\mu$), pillow lavas may well contain a submicroscopic fraction within single domain size, and owing to its relatively greater surface area, it is this fraction which will be most susceptible to later alteration. Identification of alteration rims around visible titanomagnetite grains suggests that finer grains may be entirely consumed, and destruction of this fraction may eliminate the most stable components of magnetisation residing in the rock.

Thirdly, it is also possible that stability of non-oxidised rocks depends on the strength of the magnetising field (Lawley and Ade-Hall 1971) which was possibly considerably weaker in Palaeozoic and Late Precambrian times (Smith 1967).

## 5. Palaeomagnetism of the dolerite dyke swarm

The Gwna Group pillow lavas are cut by seven impersistent dolerite dykes intruded after the Late Precambrian folding; and sampled at localities indicated in Figure 1. All samples show moderate to good stability to a.f. demagnetisation with systematic fall in moments (Fig. 5) indicating that the bulk of the magnetisation is held by a magnetite-ulvöspinel member. This indication is confirmed by petrographic examination and single Curie point thermomagnetic curves with Curie points between 550 and 580°C. Slight falls in saturation magnetisation on cooling are probably due to oxidation of the stable magnetic phase to hematite (Larson *et al.* 1968).

As a rule within-site grouping is best after demagnetising in fields of a few hundred Oersteds; initial treatment tends to remove large low coercivity components some of which are aligned close to the present Earth's field. All sites have significantly-grouped mean directions of magnetisation after treatment and fall into two groups. Sites D2, D3, D4, D5 and D6 combined to give a mean (Table II) of $D=53°$ $I=-29°$ ($k=29$), and sites D1 and D7, which with the exception of site D4 include all the most weakly magnetised samples, give a mean of $D=183°$, $I=-21{\cdot}5°$ ($k=461$). The first group gives a derived palaeomagnetic pole at 130°E, 7°N, and the second group a pole at 356°E, 48°S.

## 6. Interpretation of the palaeomagnetic pole positions

The Gwna Group pillow lavas yield a palaeomagnetic pole lying in the southwest Indian Ocean. The magnetic and petrologic data discussed in section 4 do not provide

Fig. 5. Directional behaviour, demagnetisation curves and thermomagnetic curves for samples from the dolerite dyke swarm at Llanddwyn Island. Abscissa of the demagnetisation curves are intervals of 100 oersteds peak demagnetising field and of the thermomagnetic curves are intervals of 100°C.

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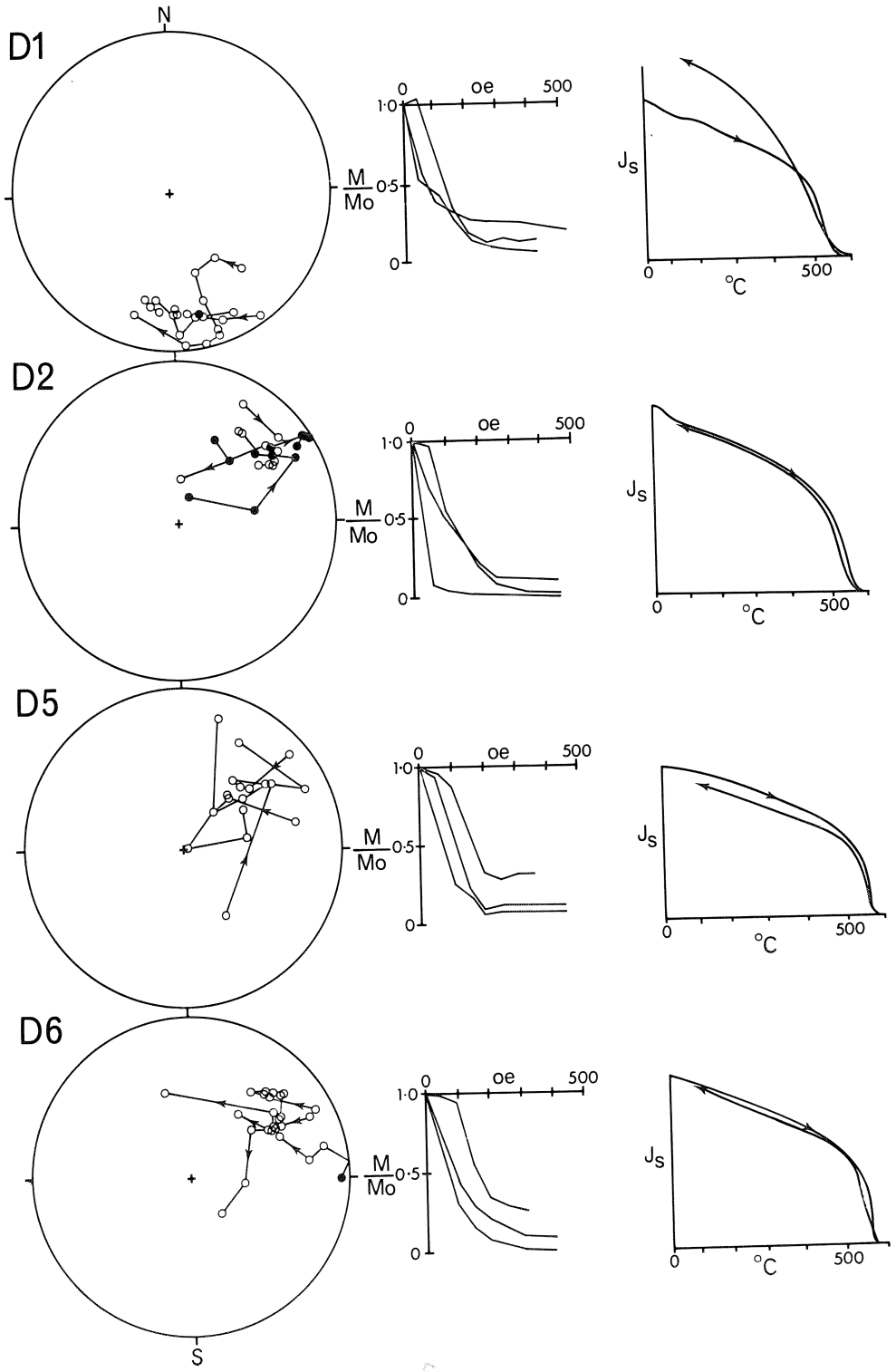


Table 2. Palaeomagnetic results from dolerite dykes, Llanddwyn Island, Anglesey (355·5°E, 53·1°N).

| | | *Before a.f. cleaning* | | | | | *After a.f. cleaning* | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | *Cleaning Fields* | | | | | *Pole Position* | |
| *Site No.* | *N* | *D* | *I* | *R* | *k* | *(Oe)* | *D* | *I* | *R* | *k* | *°E* | *°N* |
| D1 | 7 | — | — | — | — | 150–250 | 184 | −24 | 5·92 | 6 | 350 | −49 |
| D2 | 5 | 41 | 38 | 4·04 | 4 | 100–250 | 54 | −14 | 4·25 | 5 | 119 | 15 |
| D3 | 6 | 95 | 31 | 4·93 | 5 | 300–600 | 58 | −17 | 5·63 | 13 | 117 | 11 |
| D4 | 6 | 14 | 69 | 5·90 | 53 | 100–150 | 47 | −48 | 6·44 | 11 | 136 | 17 |
| D5 | 6 | 60 | −32 | 4·15 | 3 | 100–150 | 49 | −40 | 5·93 | 69 | 132 | 3 |
| D6 | 6 | 65 | −3 | 5·58 | 12 | 100–200 | 55 | −28 | 5·91 | 57 | 122 | 7 |
| D7 | 5 | — | — | — | — | 150–400 | 182 | −19 | 4·94 | 62 | 353 | −47 |
| Mean direction and pole position after a.f. cleaning: | | (i) Sites D2–D6 (N = 5) | | | | | 53·1 | −29·4 | 4·86 | 29 | 129·7 ($\alpha$ = 9·7°) | 6·9 |
| | | (ii) Sites D1 and D7 (N = 2) | | | | | 183·0 | −21·5 | 1·99 | 461 | 355·6 ($\alpha_{95}$ = 7·4°) | −48·0 |

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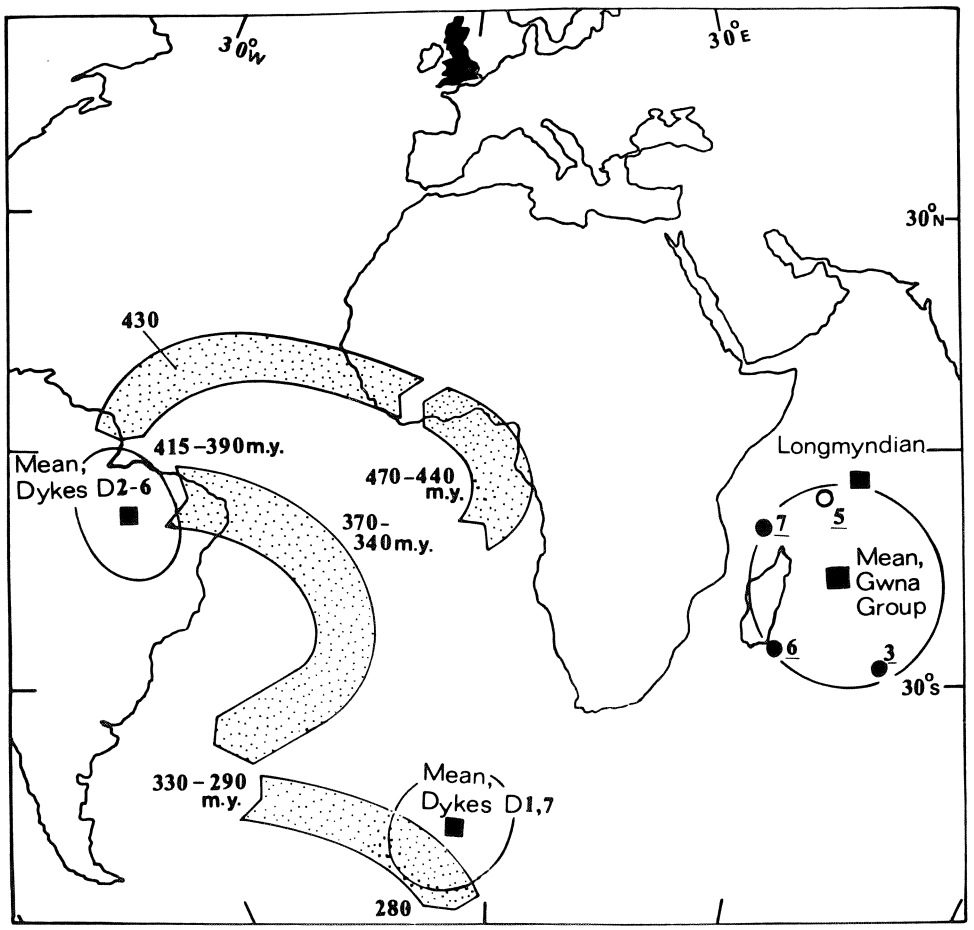



Fig. 6. The palaeomagnetic pole positions from this study plotted on an equal area map showing the apparent polar wander path for Britain from Ordovician times as a stippled swathe. The numbers refer to ages assigned to each segment of this curve in millions of years. The circles of 95% are given for each palaeomagnetic pole, and virtual geomagnetic poles are given (numbers underlined) for each site from the Monian pillow lavas.

conclusive evidence that the magnetisation of these rocks in syngenetic, but presence of a close reversal of magnetisation and the remoteness of this pole from any post-Cambrian pole from Britain are strong indications that the magnetisation dates from time of formation of the Gwna Group, and hence that this pole position relates to *c.* 600 m.y. Because of the possibility that the tectonic dip correction is not complete the quoted pole position is regarded as tentative, but the palaeomagnetic inclination of 9° does imply that this part of the Gwna Group was deposited at a palaeolatitude of about 5°. The only published pole with which to compare this result is that obtained by Creer (1957) from Longmyndian sediments of the Welsh Borderland and recently dated by Bath (1974) as having a maximum age of 600 m.y. These two pole positions are not significantly different from one another (Fig. 6). No simple apparent polar wander path is drawn in Figure 6 between these *c.* 600 m.y. pole positions and Ordovician pole positions from which they are separated by about 50° because

large amounts of apparent polar wandering appear to have characterised Eocambrian and Cambrian times (McElhinny *et al.* 1974), and have yet to be precisely defined.

The pole position from the first group of dykes lies close to the apparent polar wander curve from Britain as defined by poles of Late Silurian to Lower Devonian age (Briden *et al.* 1973 and Fig. 6). This confirms the views of Greenly (1919) that they are Palaeozoic in age, and are hence unrelated to dykes of comparable trend but Tertiary age elsewhere in Anglesey. It is noteworthy that the quite different direction of magnetisation obtained from dykes D1 and D7 is close to the Group B direction of magnetisation isolated from Lower Devonian volcanics of the Midland Valley of Scotland (Sallomy and Piper 1973). This is one of two intermediate directions of magnetisation in the latter region which are clearly related by their stratigraphic relationship to transitions of magnetic field polarity between the normal and reversed modes. They are possibly related to the equatorial dipoles which have "locked" the magnetic field direction for short periods during Tertiary to Recent polarity reversals, (Dagley and Lawley 1974).

A second possibility is that these two dykes are Permian or Triassic in age since they yield pole positions falling near to other poles from Britain of this age (e.g. Irving 1964). This is thought to be less likely however in view of their association with, and their similarity to, the remaining dykes, and their weak magnetisations; also, volcanism of this age has not yet been identified in this general region.

*Acknowledgments.* Some of the work on the pillow lavas was undertaken by D. W. McMaster, and we are very grateful to the Nature Conservancy Council for granting us permission to undertake the study at their Newborough Warren Reserve, and the University of Liverpool for field support. Dr. G. C. Brown provided useful geological advice, and Mrs. M. Catterall kindly made the thermomagnetic determinations.

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